

Assignment -02

Task 01: Please conduct multiple Protein-Ligand Docking and provide the top 10 results in the following table.

	Ligand	Binding Affinity	rmsd/ub	rmsd/lb
01	EC5e1s_5281855_uff_E=227.58	-8.6	0	0
02	EC5e1s_453214_uff_E=631.82	-7	0	0
03	EC5e1s_445858_uff_E=177.42	-5.8	0	0
04	EC5e1s_637542_uff_E=90.83	-5.7	0	0
05	EC5e1s_338_uff_E=73.50	-5.1	0	0
06	EC5e1s_72_uff_E=70.72	-5.1	0	0
07	EC5e1s_7478_uff_E=80.31	-5.1	0	0
08	EC5e1s_8655_uff_E=168.50	-5.1	0	0
09	EC5e1s_11635_uff_E=56.94	-4.5	0	0
10	EC5e1s_72_uff_E=70.72	-5.1	2.26	1.034

Task 02: Please perform ADME analysis on the 10 compounds obtained from the Protein-Ligand Docking and provide the results in the table below.

ADME Analysis								Pharmacokinetics		Drug likeness	Medicinal Chemistry
Name	CID ID	Canonical SMILES	Molecular weight	Num. H-bond acceptors	Num. H-bond donors	Lipophilicity (iLOGP)	Water Solubility (Log S (SILICOS-IT))	GI absorption	BBB permeant	Lipinski	PAINS
01	453214	<chem>COC(=O)C1=CO[C@H]([C@H]2[C@@H]1C=C[C@@]23C=C(C(=O)O3)CO)O[C@H]4[C@@H]([C@H]([C@@H]([C@H](O4)CO)O)O</chem>	456.40 g/mol	12	5	2.14	1.77	low	no	Yes; 1 violation: NorO >10	0 alert

02	528 185 5	<chem>C1=C2C3=C(C(=C1O)O)OC(=O)C4=CC(=C(C(=C43)OC2=O)O)O</chem>	302.19 g/mol	8	4	0.79	-3.35	high	no	Yes; 0 violat ion	1 alert: catech ol_A
03	865 5	<chem>COC1=CC(=CC(=C1O)OC)C=O</chem>	182.17 g/mol	4	1	1.66	-2.03	high	yes	Yes; 0 violat ion	0 alert
04	135	<chem>C1=CC(=CC(=C1C(=O)O)O</chem>	138.12 g/mol	3	2	0.85	-1.17	high	yes	Yes; 0 violat ion	0 alert
05	747 8	<chem>COC1=CC=C(C(=C1)C(=O)O</chem>	152.15 g/mol	3	1	1.56	-1.88	high	yes	Yes; 0 violat ion	0 alert
06	338	<chem>C1=CC=C(C(C(=C1)C(=O)O)O</chem>	138.12 g/mol	3	2	1.13	-1.17	high	yes	Yes; 0 violat ion	0 alert
07	445 858	<chem>COC1=C(C(=CC(=C1)/C=C/C(=O)O)O</chem>	194.18 g/mol	4	2	1.62	-1.42	high	yes	Yes; 0 violat ion	0 alert
08	72	<chem>C1=CC(=C(C(=C1C(=O)O)O)O</chem>	154.12 g/mol	4	3	0.66	-0.60	high	no	Yes; 0 violat ion	1 alert: catech ol_A
09	637 542	<chem>C1=CC(=CC(=C1/C=C/C(=O)O)O</chem>	164.16 g/mol	3	2	0.95	-1.28	high	yes	Yes; 0 violat ion	0 alert
10	445 858	<chem>COC1=C(C(=CC(=C1)/C=C/C(=O)O)O</chem>	194.18 g/mol	4	2	1.62	-1.42	high	yes	Yes; 0 violat ion	0 alert

Task 03: Perform Toxicity Prediction on the 10 compounds obtained from the Protein-Ligand Docking and provide the results in the table below.

Toxicity Prediction							
Name	CID ID	Canonical SMILES	Hepatotoxicity	Carcinogenicity	Immunotoxicity	Mutagenicity	Cytotoxicity
01	453214	<chem>COC(=O)C1=CO[C@H]([C@H]2[C@@H]1C=C[C@@]23C=C(C(=O)O3)CO)O[C@H]4[C@@H]([C@H]([C@@H]([C@H](O4)CO)O)O)O</chem>	Inactive	Inactive	Active	Inactive	Inactive
02	5281855	<chem>C1=C2C3=C(C(=C1O)O)OC(=O)C4=CC(=C(C(=C43)OC2=O)O)O</chem>	Inactive	Active	Inactive	Inactive	Inactive
03	8655	<chem>COC1=CC(=CC(=C1O)OC)C=O</chem>	Inactive	Inactive	Inactive	Inactive	Inactive
04	135	<chem>C1=CC(=CC=C1C(=O)O)O</chem>	Inactive	Inactive	Inactive	Inactive	Inactive
05	7478	<chem>COC1=CC=C(C(=C1)C(=O)O</chem>	Active	Inactive	Inactive	Inactive	Inactive
06	338	<chem>C1=CC=C(C(=C1)C(=O)O)O</chem>	Active	Inactive	Inactive	Inactive	Inactive
07	445858	<chem>COC1=C(C=CC(=C1)/C=C/C(=O)O)O</chem>	Inactive	Inactive	Active	Inactive	Inactive
08	72	<chem>C1=CC(=C(C(=C1C(=O)O)O)O)O</chem>	Inactive	Active	Inactive	Inactive	Inactive
09	637542	<chem>C1=CC(=CC=C1/C=C/C(=O)O)O</chem>	Active	Inactive	Active	Inactive	Inactive
10	445858	<chem>COC1=C(C=CC(=C1)/C=C/C(=O)O)O</chem>	Inactive	Inactive	Active	Inactive	Inactive

Task 04: Identify the highest-ranking Protein – ligand complex and input the corresponding figures into the table below.

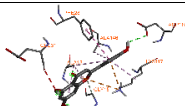
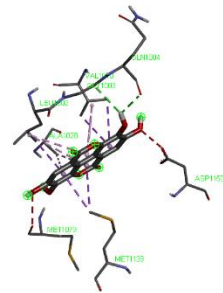
Figure Name	Sample Figure	Input your Docking Figure
Figure 01		

Figure 02

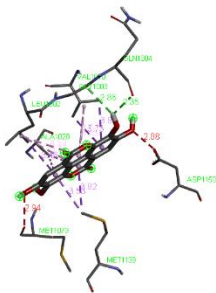
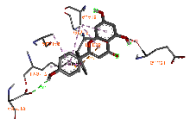


Figure 03

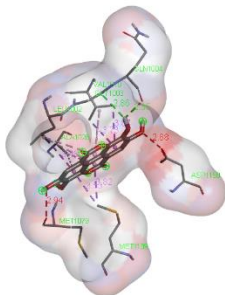
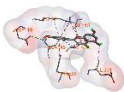


Figure 04

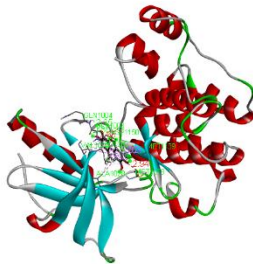
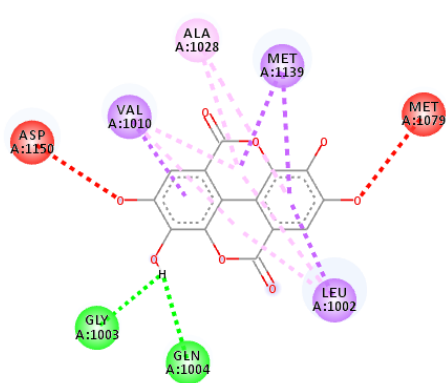
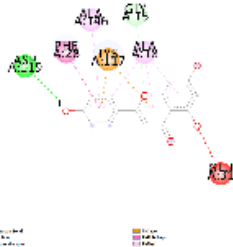


Figure 05



Interactions

Conventional Hydrogen Bond	Pi-Sigma
Unfavorable Acceptor-Acceptor	Pi-Alkyl

Task 05: Identify the highest-ranking Protein – ligand complex and input the Interaction details into the table below.

Name	Distance	Category	Types
N:UNK1:H - A:GLY1003:O	2.86292	Hydrogen Bond	Conventional Hydrogen Bond
N:UNK1:H - A:GLN1004:O	2.34832	Hydrogen Bond	Conventional Hydrogen Bond
A:LEU1002:CD1 - N:UNK1	3.46423	Hydrophobic	Pi-Alkyl
A:VAL1010:CG1 - N:UNK1	3.73158	Hydrophobic	Pi-Alkyl
A:VAL1010:CG2 - N:UNK1	3.87256	Hydrophobic	Pi-Alkyl
A:MET1139:CE - N:UNK1	3.52589	Hydrophobic	Pi-Alkyl
A:MET1139:CE - N:UNK1	3.82239	Hydrophobic	Pi-Alkyl
N:UNK1 - A:LEU1002	4.59424	Hydrophobic	Pi-Alkyl
N:UNK1 - A:LEU1002	5.21322	Hydrophobic	Pi-Sigma
N:UNK1 - A:VAL1010	4.73698	Hydrophobic	Pi-Sigma
N:UNK1 - A:VAL1010	4.72553	Hydrophobic	Pi-Sigma
N:UNK1 - A:ALA1028	5.29282	Hydrophobic	Pi-Sigma
N:UNK1 - A:ALA1028	4.37613	Hydrophobic	Pi-Sigma